Identification and characterization of key long non-coding RNAs involved in cervical cancer progression using a systems biology approach

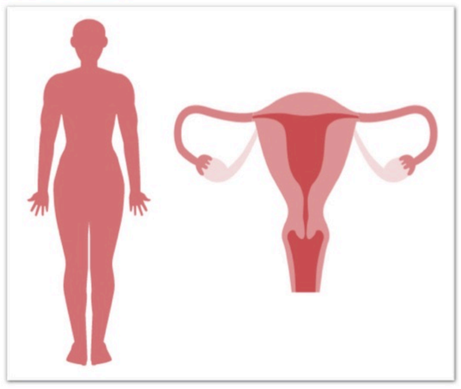
Mallikarjuna Thippana[1], Vaibhav Vindal[1*]

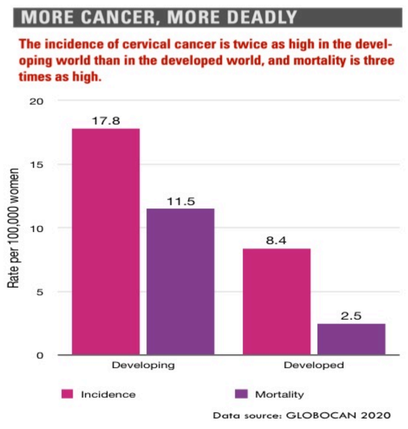
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Introduction





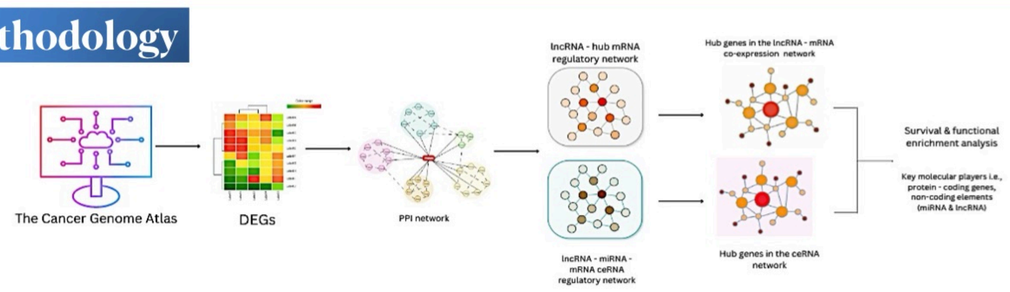
Cervical cancer (CC) is the primary cause of mortality among women in developing countries. It is estimated that 604,127 cervical cancer cases have been reported in 2020 globally. In India, the incident and mortality rates are 14.7 and 9.2, respectively, per 100,000 women. Preventing cervical cancer is partially possible by early vaccination against the human papillomavirus, the most common cause of the disease. Nevertheless, it is imperative to understand the genetics of the disease progression to develop new therapeutic strategies.

Aims & Objectives

To identify the novel non-coding elements that has a role in cervical cancer progression, we explored an integrative network approach with the following objectives.

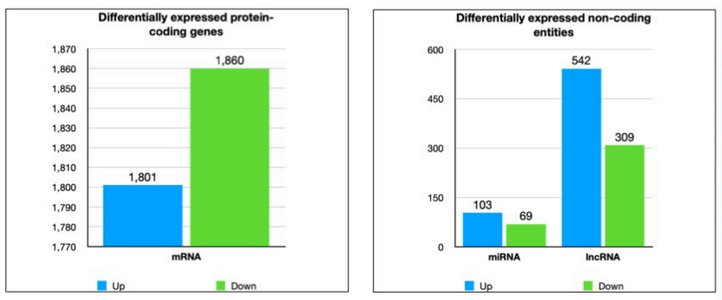
1. Screening significantly differentially expressed genes (protein-coding and non-coding)
2. Identifying significant hubs from protein-protein interaction networks
3. Identifying key ncRNAs and their target genes from regulatory networks
4. Web resource development to host findings from the study

Methodology



Differential expression analysis

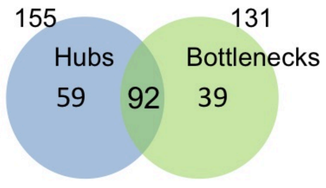
To identify significantly differentially expressed protein-coding genes and ncRNAs (miRNA & lncRNA) associated with cervical cancer progression we utilized RNA-seq data available at TCGA GDC data portal along with GTEx normal sample expression data.

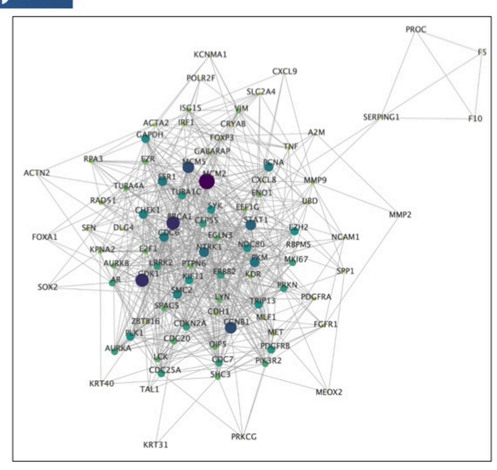


Protein-protein interaction network analysis

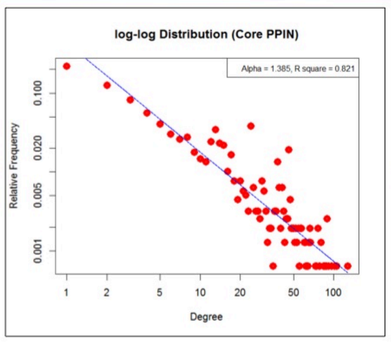
The physical interactions between the identified DEG's were investigated by dysregulated protein-protein interaction network analysis.

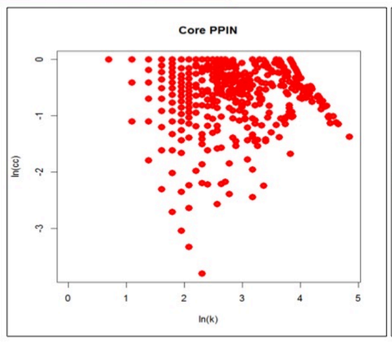
No. of nodes	No. edges
2487	11574

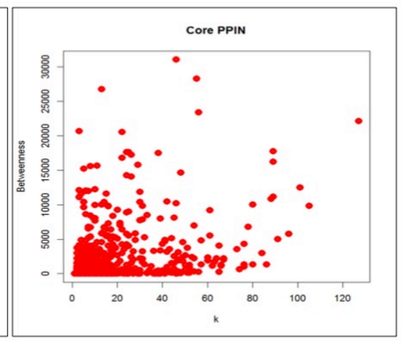




Topological properties of networks





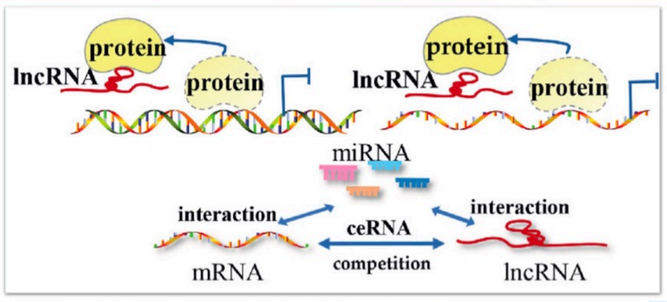


Left panel: Degree distribution of core or primary network and Turquoise module networks follow the property of power law and scale free nature.

middle panel: Degree and clustering coefficient (CC) of both core and turquoise module networks are negatively correlated.

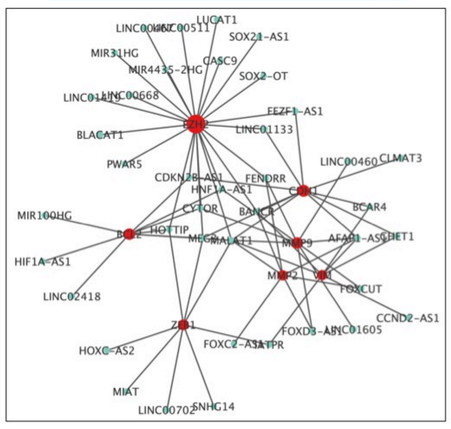
Right panel: Degree and Betweenness correlations of core and turquoise module networks are positively correlated.

lncRNA-mRNA co-expression network analysis



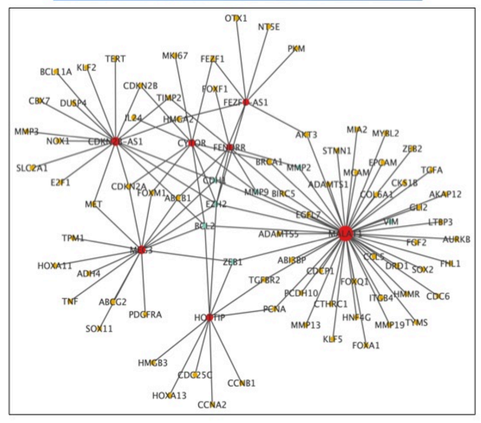
1. lncRNA-mRNA interactions
   LncTarD database
2. lncRNA-miRNA interactions
   miRcode database
3. miRNA-mRNA interactions
   miRdb database
   TargetScan database
   miRTarBase database

Indegree of network



EZH2 CDH1 BCL2 MMP9 ZEB1 MMP2 VIM

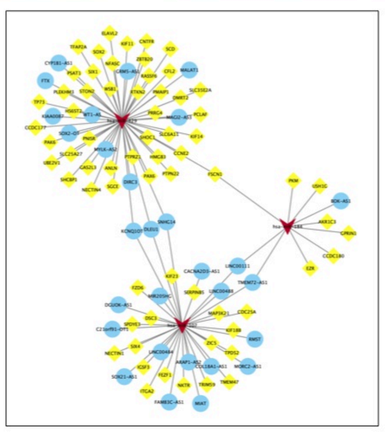
outdegree of network



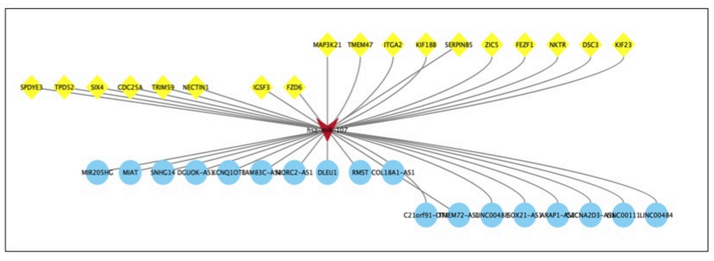
MALAT1 CDKN2B-AS1 MEG3 HOTTIP CYTOR FEZF1-AS1 FENDRR

lncRNA	Gene (mRNA)	Binding site	Binding energy (kcal/mol)
MALAT1(476-531)	EZH2(865-920)	CDS	-473.12
MALAT1(467-521)	CDH1(2967-3021)	3'UTR	-778.64
MALAT1(465-516)	BCL2(1720-1771)	CDS	-1490.75
MALAT1(444-495)	ZEB1(6117-6168)	3'UTR	-1624.98
MALAT1(470-525)	MMP2(3112-3167)	3'UTR	-538.78
MALAT1(759-808)	VIN(1301-1350)	CDS	-210.26
MEG3(258-305)	ZEB1(4624-4671)	3'UTR	-216.63
MEG3(596-632)	BCL2(6161-6212)	3'UTR	-424.78
MEG3(449-498)	CDH1(4501-4550)	3'UTR	-246.24
MEG3(403-450)	MET(504-551)	CDS	-320.18

lncRNA-miRNA-mRNA ceRNA network analysis

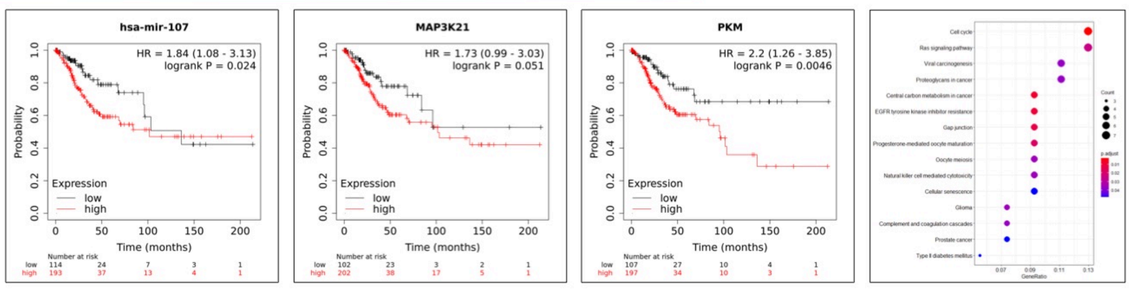


lncRNA-miRNA-mRNA interaction network (color representation: Blue: lncRNA; Darkred: miRNA; Yellow: mRNA)



lncRNAs *KCNQ1OT1, SNHG14, DLEU1* sponging miRNA *hsa-miR-107* to regulate coding genes

Survival & Functional Enrichment analysis



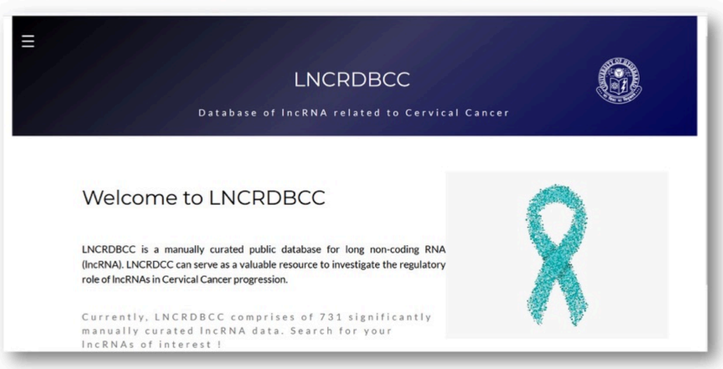
Prognostic performance genes via survival analysis

KEGG pathways

LNCRDBCC: Database of lncRNA related to Cervical Cancer

The findings reported from the study were hosted in the web resource at Bioinformatics facility (BIF) at University of Hyderabad, Hyderabad, India

Scan code to access the database



Acknowledgements

Laboratory of Computational Functional Genomics
School of Life Sciences University Of Hyderabad

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